Answer Key - Mathematics - K-12 Level Assessment 9 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
Options: A) 800; B) 830; C) 840; D) 870
Note: The correct answer is 840 because when rounding to the nearest ten, you look at the digit in the ones place (6 in this case), and since it is 5 or greater, you round up the tens digit from 3 to 4, resulting in 840.

2. **Answer:** C
Options: A) $\frac{\pi}{24}$; B) $\frac{\pi}{12}$; C) $\frac{\pi}{6}$; D) $\frac{\pi}{3}$
Note: The correct answer is $\frac{\pi}{6}$ because the region defined by $x^2+y^2+z^2 \leq 1$ represents a sphere of radius 1, and the volume of this sphere, when compared to the volume of the cube defined by the constraints on x, y, and z, gives the probability as the ratio of the sphere's volume $\frac{4}{3}\pi$ to the cube's volume 8, simplifying to $\frac{\pi}{6}$.

3. **Answer:** A
Options: A) $\frac{8}{45}$; B) $\frac{8}{5}$; C) $\frac{8}{7}$; D) $\frac{8}{33}$
Note: To convert the repeating decimal $0.1\overline{7}$ into a fraction, we let $x = 0.1\overline{7}$, multiply by 10 to shift the decimal, and then subtract the resulting equations to isolate the repeating part, ultimately leading to the fraction $\frac{8}{45}$ after simplification.

4. **Answer:** D
Options: A) $\frac{7}{2}$; B) $\frac{7}{9}$; C) $\frac{9}{2}$; D) $\frac{9}{7}$
Note: The reciprocal of a number is found by dividing 1 by that number, and since $0.\overline{7}$ equals $\frac{7}{9}$, its reciprocal is $\frac{1}{\frac{7}{9}} = \frac{9}{7}$.

5. **Answer:** B
Options: A) 6; B) 12; C) 96; D) 102
Note: The integer 12, when its digits are switched to form 21 and then adding 3 gives 24, which is equal to 2 times 12, satisfying the condition of the problem.

True / False

6. **Answer:** False
Options: A) True; B) False
Note: The statement is false because -4 plus -3 equals -7, not -1.

7. **Answer:** False
Options: A) True; B) False
Note: The 5 th term in the series, which starts with 2 and multiplies by -2 each time, is actually 32, not -16, as the sequence alternates between positive and negative values while doubling in magnitude.

8. **Answer:** False
Options: A) True; B) False
Note: The median of the data set is actually 22, as the numbers must be arranged in ascending order and the middle value identified, which shows that the statement claiming the median is 20 is incorrect.

9. **Answer:** False
Options: A) True; B) False
Note: A simple random sample is defined by the method of selection, where every individual in the population has an equal chance of being chosen, rather than how representative the sample is of the population.

10. **Answer:** False
Options: A) True; B) False
Note: The statement is false because if 7 coins represent 5% of Kate's collection, then her total collection is actually 140 coins, not 120.

Long Form Response

11. **Answer:** There are six ways in which one European country can be chosen, four in which one Asian country can be chosen, three in which one North American country can be chosen, and seven in which one African country can be chosen. Thus, there are $6 \cdot 4 \cdot 3 \cdot 7 = 504$ ways to form the international commission.
Note: correct because it applies the fundamental principle of counting, which states that the total number of combinations can be found by multiplying the number of choices for each category, leading to $6 \cdot 4 \cdot 3 \cdot 7 = 504$ ways to form the international commission.

12. **Answer:** Increasing the sample size from 200 to 2,000 in a study of geese returning to the same breeding site will affect the distribution of the sample proportion by making it less spread out. This means that the estimates we get from our sample will be more accurate and closer to the true proportion of geese that return to the breeding site. A larger sample size reduces the variability in the results because it provides a better representation of the entire population. As a result, the distribution of the sample proportion becomes narrower, leading to more reliable conclusions about the geese's behavior.
Note: Increasing the sample size reduces variability and narrows the distribution of the sample proportion, leading to more accurate estimates that better reflect the true population parameter.

End of Answer Key